Voice-Based Parkinson's Disease Classification Using Handcrafted Acoustic Features and Log-Mel Spectrograms

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Abstract

Parkinson's disease (PD) can affect speech production, motivating research into voice recordings as a non-invasive source of acoustic information. This study compares a classical machine-learning pipeline based on handcrafted acoustic summaries with a convolutional neural network (CNN) based on Log-Mel spectrograms. The experiments use an Italian Parkinson's Voice and Speech collection containing 831 WAV recordings from 65 participants: 37 healthy controls and 28 participants with PD. Participants, rather than individual recordings, are assigned to disjoint development and test partitions. Four classical classifiers are evaluated using five-fold participant-stratified cross-validation, and an RBF-kernel support vector machine (SVM) is selected. A separate CNN processes five-second, 64-band Log-Mel inputs. On 226 held-out recordings from 17 participants, the SVM achieves 0.9558 recording-level accuracy and 0.9412 participant-level accuracy; the CNN achieves 0.9690 and 0.9412, respectively. The SVM has a higher recording-level ROC-AUC, while both models have identical participant-level metrics in the saved run. The small held-out cohort and absence of external validation limit generalisation. This work is an experimental research prototype, not a clinical diagnostic tool.

Keywords: Parkinson's disease; voice analysis; acoustic features; support vector machine; convolutional neural network; Log-Mel spectrogram; participant-level evaluation.

1 Introduction

Parkinson's disease (PD) is a progressive neurological disorder that can affect speech and voice production. Changes in phonation, articulation, timing, and prosody motivate research into acoustic measurements as potential indicators of motor and speech impairment. Voice analysis is attractive as a non-invasive and comparatively accessible research modality; however, a computational classifier trained on a limited dataset should not be interpreted as a clinical diagnostic instrument.

A central methodological issue is the repeated-measures structure of speech datasets. Multiple recordings from one person are correlated and may share speaker-specific characteristics. If recordings from the same participant are divided between training and testing, a model may exploit person-specific cues and produce an optimistic estimate of performance on unseen people. The present study therefore separates participants before model development and reports both recording-level and participant-level metrics.

The research question is: how do a classical machine-learning pipeline based on statistical summaries of handcrafted acoustic features and a convolutional neural network (CNN) based on Log-Mel spectrograms compare when tested on participants excluded from model development? The objectives are to audit the available cohort, implement both representations, compare classical classifiers using participant-grouped cross-validation, and evaluate the selected models on the same held-out participants.

The contribution is an implementation-level comparison on the Italian Parkinson's Voice and Speech collection using participant-disjoint evaluation, dual-level reporting, and exploratory feature-importance analysis. The study does not claim a new clinical biomarker or superiority over published methods.

2 Related Work

Research on voice-based PD classification has followed two broad directions: handcrafted acoustic descriptors paired with conventional classifiers, and learned representations processed by neural networks. Reviews by Ngo *et al.* and Amato *et al.* describe a diverse literature spanning speech tasks, signal descriptors, and statistical or machine-learning classifiers [2, 3]. These reviews also make clear that studies differ in their datasets, recording protocols, and evaluation procedures, limiting direct comparison of headline accuracy values.

Handcrafted pipelines commonly summarise measurable properties of the voice signal and use classifiers such as support vector machines (SVMs), random forests, or neural networks. Their advantage is that the input variables have interpretable acoustic definitions and can be inspected using feature analysis. Their limitation is that the selected descriptors may not preserve all temporal and contextual information in a speech signal. Amato *et al.* survey feature-based machine-learning methods and discuss the role of acoustic features in PD voice analysis [3]. In the present work, MFCC, delta-MFCC, spectral, energy, and chroma summaries form the tabular representation, with SVM and tree-based classifiers used as comparators.

A second direction uses neural networks to learn discriminative patterns from structured signal representations. A Log-Mel spectrogram retains local time-frequency structure, allowing convolutional filters to learn patterns across neighbouring time and frequency regions. Such learned representations are flexible, but their performance depends on data volume, architecture, recording conditions, and the evaluation protocol. The current study uses a compact CNN as a controlled alternative to the handcrafted-feature pipeline rather than claiming that deep learning is inherently superior.

Luna-Ortiz *et al.* proposed an improved smallest normalized difference associative memory (ISNDAM) method, incorporating a feature-relevance identification phase, and compared it with a large set of classifiers on two established voice datasets [1]. Their paper is the supplied base reference for this project. The method in the present notebook is different: it implements Logistic Regression, RBF-SVM, Random Forest, Extra Trees, and a Log-Mel CNN. ISNDAM is not implemented here, and the published ISNDAM results are not results of this project.

Recent reviews reinforce the need to interpret high reported performance carefully. Ngo *et al.* reviewed computerized speech and voice analysis and highlighted the variation in datasets and analysis methods [2]. Xavier *et al.* reviewed voice analysis studies published between 2019 and 2023, showing the breadth of approaches and tasks in the field [4]. A 2025 systematic review by Farajollahi *et al.* similarly describes a field with many reported high-performing models but persistent concerns around dataset heterogeneity, validation, calibration, and clinical

Table 1: Selected literature and its relationship to the present implementation

Reference	Research focus	Relevance and comparison boundary
Ngo *et al.* (2022) [2]	Systematic review of computerized speech and voice analysis	Synthesises datasets, protocols, signal analysis, and features; highlights heterogeneity that complicates direct performance comparison.
Amato *et al.* (2023) [3]	Survey of machine-learning and statistical voice analysis	Provides context for handcrafted acoustic descriptors and conventional classifiers, including the feature-based branch used here.
Luna-Ortiz *et al.* (2023) [1]	ISNDAM associative-memory classifier on two established voice datasets	Supplied base paper; method differs from the SVM and CNN implemented in this project.
Xavier *et al.* (2025) [4]	Systematic review of voice-analysis studies from 2019–2023	Demonstrates breadth of research tasks and methods; supports cautious interpretation across non-matched datasets.
Farajollahi *et al.* (2025) [5]	Systematic review of ML/DL approaches	Discusses dataset heterogeneity, validation, and clinical-translation limitations relevant to this experiment.
Hossain *et al.* (2026) [6]	Machine-learning analysis of the IPVS dataset	Uses the same named 65-participant collection; provides dataset-specific context, not a direct benchmark for our split or metrics.

translation [5]. These reviews motivate transparent reporting of the participant split and the unit of evaluation.

The dataset used here is the Italian Parkinson's Voice and Speech (IPVS) collection. Hossain, Traini, and Amenta describe the IPVS dataset as comprising 65 native Italian speakers divided into 15 young healthy controls, 22 elderly healthy controls, and 28 participants with PD [6]. This matches the cohort counts found in the project notebook. Their work is particularly relevant because it concerns the same named dataset; however, its reported methods and outcomes must remain distinct from the experiments in this paper.

2.1 Research Gap and Positioning

The literature does not support claiming that voice-based PD classification is unexplored. Rather, the practical question addressed here is narrower: within the available IPVS project data, how do a compact handcrafted-feature pipeline and a compact Log-Mel CNN compare when participants are kept disjoint between model development and final testing? The paper's value lies in documenting this implementation and reporting both file-level and participant-level outcomes. A fair comparison with other publications would require matching cohorts, speech tasks, preprocessing, participant partitions, and evaluation units; such a matched cross-study experiment is outside the scope of this project.

3 Dataset Description

The notebook audits three cohorts: young healthy controls (YHC), elderly healthy controls (EHC), and participants with PD. The corrected audit identifies 65 unique participants and 831 WAV recordings. The two healthy-control cohorts are combined into the binary healthy-control (HC) class. At recording level, there are 394 HC and 437 PD recordings; at participant level, there are 37 HC and 28 PD participants.

The directory scanner accounts for the different folder structures used by the cohorts. For the

Table 2: Participant and recording distribution from the notebook audit

Cohort	Participants	Recordings
Young healthy controls	15	45
Elderly healthy controls	22	349
Parkinson's disease	28	437
Total	65	831

PD group, the range directory is retained in the participant identifier to avoid collisions between repeated folder names. Participant identifiers are used for grouping and auditing only; they are not supplied to the classifiers as predictive variables. The saved audit reports no unknown task codes and no feature-extraction errors.

4 Methodology

4.1 Overall Workflow and Participant Split

Fig. 1 summarises the implemented workflow. After cohort and participant auditing, the participant table is stratified into training and test participants. All recordings from a participant remain in the same partition. The saved split contains 48 training participants (27 HC and 21 PD) and 17 held-out test participants (10 HC and 7 PD), corresponding to 605 training recordings and 226 test recordings. The held-out participants are not used for model selection.

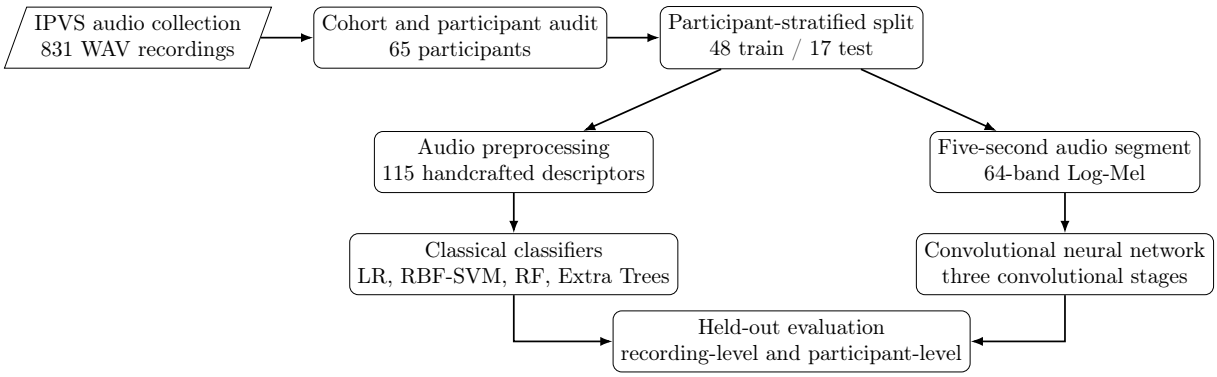


Figure 1: Team-developed schematic of the two implemented audio-classification branches.

For classical models, five-fold stratified cross-validation is conducted using training participants only. Each fold is formed at participant level, and all recordings from the fold's participants are mapped to the corresponding training or validation subset. Balanced accuracy is used to compare classifiers. This prevents participant overlap between folds, although the reported fold metric is calculated over the recordings belonging to each fold.

4.2 Audio Preprocessing and Handcrafted Features

Audio is loaded from WAV files and converted to mono by averaging channels when necessary. Signals are resampled to 16 kHz, mean-centred, and peak-normalised when the peak amplitude is nonzero. For a discrete recording $x[n]$, mean-centering is

$$x_c[n] = x[n] - \bar{x}, \tag{1}$$

Table 3: Five-fold participant-stratified cross-validation results

Classifier	Mean BA	Std. dev.
RBF-SVM	0.9606	0.0360
Extra Trees	0.9483	0.0474
Random Forest	0.9404	0.0469
Logistic Regression	0.9328	0.0302

and peak normalisation is

$$x_p[n] = \frac{x_c[n]}{\max_n |x_c[n]|}. \tag{2}$$

The normalisation scales the maximum absolute sample; it does not remove differences in microphone, room, or speaking conditions.

The notebook extracts 20 MFCC coefficients and their delta descriptors. It additionally computes spectral centroid, spectral bandwidth, spectral rolloff, zero-crossing rate (ZCR), root-mean-square (RMS) energy, and chroma features. For each frame-based feature sequence, the mean and standard deviation are calculated. Together with the duration feature, these operations produce 115 feature columns per recording.

MFCCs summarise the short-time spectral envelope. The signal is framed and windowed, transformed using the short-time Fourier transform, mapped through Mel-spaced filters, logarithmically compressed, and decorrelated using a discrete cosine transform. Delta-MFCCs describe temporal changes in the cepstral sequence. Spectral centroid estimates the weighted centre of spectral energy; bandwidth measures its spread; rolloff identifies a frequency below which a specified proportion of energy lies; ZCR counts sign changes; RMS summarises frame energy; and chroma maps spectral energy into pitch-class bins.

For a frame-level sequence $z_1, \ldots, z_T$, the summary statistics are

$$\mu_z = \frac{1}{T}\sum_{t=1}^{T} z_t, \qquad \sigma_z = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(z_t - \mu_z)^2}. \tag{3}$$

The fixed-length summaries permit conventional classifiers to process recordings with different durations.

4.3 Classical Machine-Learning Models

Four classifiers are evaluated: Logistic Regression, an RBF-kernel SVM, Random Forest, and Extra Trees. Median imputation and standardisation are included in the Logistic Regression and SVM pipelines. The tree-based pipelines use median imputation without standardisation. Class weighting is enabled. The SVM is selected because it obtains the highest mean balanced accuracy in the five training-participant folds.

4.4 Log-Mel Spectrogram and CNN

For the CNN branch, each recording is resampled to 16 kHz and represented by a five-second segment taken from the beginning of the recording. Shorter signals are zero-padded and longer signals are truncated. The Mel spectrogram uses 64 Mel bands, an FFT size of 1024, and a hop length of 256 samples. Power values are converted to decibels relative to the maximum value.

Table 4: CNN configuration

Component	Configuration
Input	$64 \times 313 \times 1$
Convolution filters	16, 32, 64
Kernel size	3×3
Pooling	Max pooling after stages 1 and 2
Aggregation	Global average pooling
Dense layer	32 units, ReLU
Dropout	0.45 and 0.35
Output	1 unit, sigmoid
Optimizer	Adam, 3×10^{-4}
Loss	Binary cross-entropy
Batch / maximum epochs	16 / 60

Each input has shape $64 \times 313 \times 1$.

The CNN contains three convolutional stages with 16, 32, and 64 filters. The first two stages are followed by batch normalisation and max pooling. The third stage is followed by global average pooling. The classification head contains dropout (0.45), a 32-unit ReLU dense layer, dropout (0.35), and a one-unit sigmoid output. The model uses Adam with learning rate 3×10^{-4} and binary cross-entropy. Training is limited to 60 epochs with batch size 16, early stopping (validation-loss patience 8), and learning-rate reduction on plateau (factor 0.5, patience 3, minimum learning rate 10^{-6}).

The 48 development participants are split into 38 training and 10 validation participants. The 17 held-out test participants remain unchanged. Sample weights are constructed so that participants contribute equally overall, followed by participant-level class balancing.

4.5 Implementation and Reproducibility

The notebook fixes the target sampling rate at 16 kHz and the handcrafted representation at 20 MFCC coefficients plus delta-MFCC and spectral, energy, and chroma descriptors. For the CNN, the input duration is five seconds, the Mel filter count is 64, the FFT size is 1024, and the hop length is 256. The saved participant split is stratified with a fixed random seed in the notebook. The classical model comparison uses a separate participant-level five-fold procedure on the training group. The CNN uses a second participant-level split of the 48 development participants into 38 training and 10 validation participants. The same 17 participants are retained as the test group.

These settings make the reported experiment traceable to the submitted notebook. Reproduction still depends on the original audio files, the exact notebook environment and package versions, the random seeds, and the saved split. The project should archive the final executed notebook, environment information, and generated output tables together with the manuscript.

4.6 Evaluation Metrics and Participant Aggregation

The study reports accuracy, balanced accuracy, PD precision, PD recall, F1-score, and ROC-AUC. Accuracy is the fraction of correct predictions. Balanced accuracy is the mean of sensitivity and specificity. Precision measures the proportion of positive predictions that are correct; recall

measures the proportion of PD cases detected; F1 is their harmonic mean; and ROC-AUC measures ranking performance across thresholds.

For participant-level evaluation, recording-level PD probabilities are averaged within each participant. If participant q has R_q recordings, the aggregate score is

$$\bar{p}_q = \frac{1}{R_q} \sum_{r=1}^{R_q} p_{qr}. \tag{4}$$

A participant is assigned to the PD class when $\bar{p}_q \geq 0.5$. This is a distinct evaluation unit from an individual recording.

5 Experimental Results

The selected RBF-SVM is evaluated on 226 held-out recordings from 17 participants. The CNN is evaluated on the same test recordings. Table 5 reports the metrics at both recording and participant levels.

Table 5: Held-out test performance for the two implemented models

Model	Evaluation unit	n	Accuracy	Balanced acc.	PD precision	PD recall	PD F1 / AUC
RBF-SVM	Recording	226	0.9558	0.9568	0.9358	0.9714	0.9533 / 0.9939
CNN	Recording	226	0.9690	0.9692	0.9623	0.9714	0.9668 / 0.9892
RBF-SVM	Participant	17	0.9412	0.9500	0.8750	1.0000	0.9333 / 1.0000
CNN	Participant	17	0.9412	0.9500	0.8750	1.0000	0.9333 / 1.0000

At recording level, the CNN has higher accuracy (0.9690 versus 0.9558), balanced accuracy (0.9692 versus 0.9568), precision, and F1-score. Both models have the same PD recall (0.9714). The SVM has a higher ROC-AUC (0.9939 versus 0.9892). Thus, the comparison does not yield a single model that dominates every metric. No statistical significance test is implemented, so these numerical differences should not be described as statistically significant.

At participant level, both models have the same reported metrics: 0.9412 accuracy, 0.9500 balanced accuracy, 0.8750 PD precision, 1.0000 PD recall, 0.9333 F1-score, and 1.0000 ROC-AUC. The participant test set contains only 17 people, so one participant corresponds to approximately 5.9 percentage points of accuracy. The 226 recordings are repeated observations from those 17 participants and must not be interpreted as 226 independent subjects.

5.1 Exploratory Feature Importance

Permutation importance is computed for the selected SVM using balanced accuracy on the held-out test recordings. The largest mean importance values in the saved output include MFCC means, chroma variability, MFCC standard deviations, delta-MFCC, spectral bandwidth, spectral rolloff, and RMS variability. Fig. 3 shows the ten highest-ranked entries in the saved output.

Permutation importance is conditional on the fitted model, dataset, and evaluation sample. Correlated descriptors may share predictive information. The ranking is exploratory and does not establish causality, a biological mechanism, or a clinically validated biomarker.

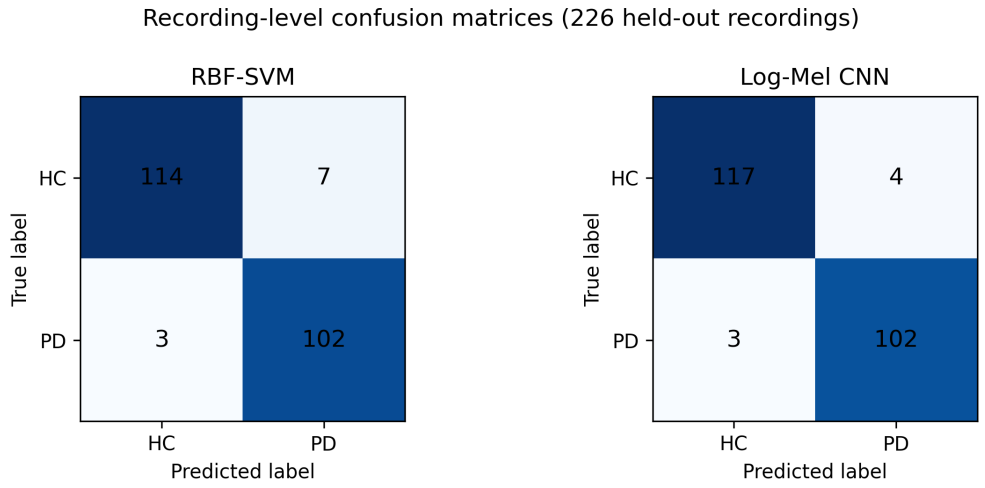


Figure 2: Recording-level confusion matrices reconstructed from the notebook's held-out classification reports. Rows indicate true classes and columns indicate predicted classes.

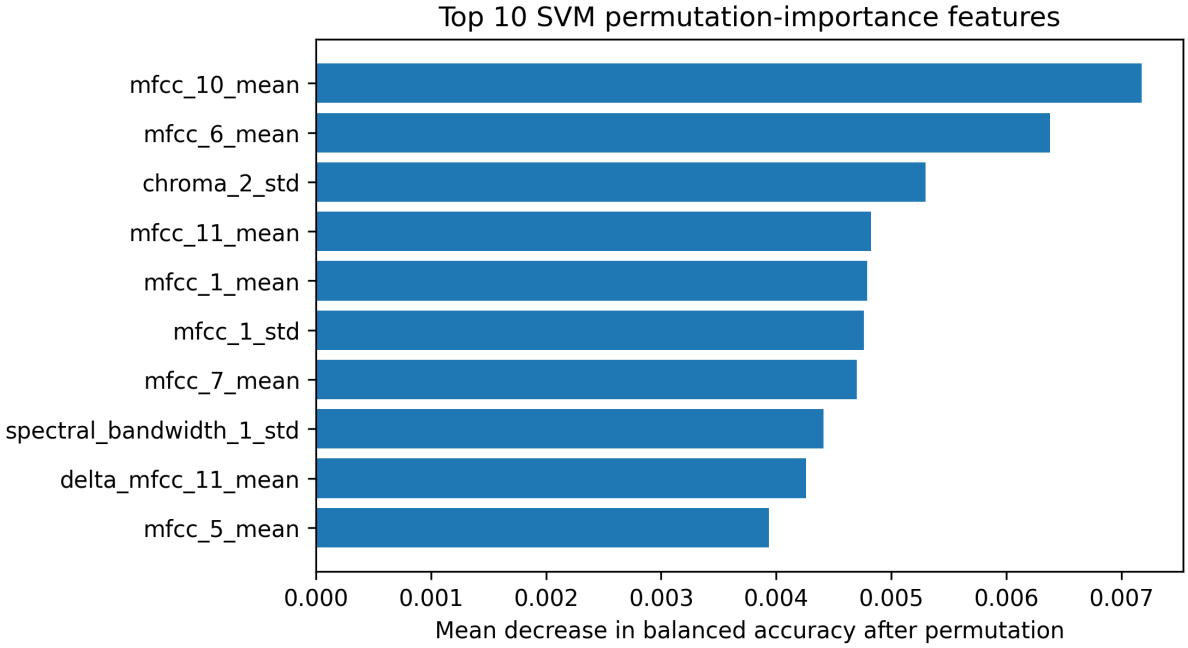


Figure 3: Top ten SVM permutation-importance features from the saved notebook output.

6 Discussion

The participant-aware split is an important design choice because it prevents recordings from the same individual from appearing in both training and held-out testing. The separate participant-level results provide a more direct view of performance on unseen people than file-level scores alone. Nevertheless, the participant test set is small, and its metrics have limited precision.

The two approaches show complementary behaviour. The CNN is stronger on several recording-level threshold metrics, while the SVM has the higher recording-level ROC-AUC. Their participant-level scores are identical in this saved run. These observations are specific to this dataset, split, and training procedure. They do not demonstrate that CNNs or SVMs are generally preferable for PD voice classification.

The study should also be interpreted in light of the related literature. Reviews describe substantial variation in datasets, language, speech tasks, feature definitions, and validation procedures [2–5]. Consequently, direct numerical comparisons with published results are not justified unless experimental conditions are matched. The ISNDAM paper is relevant background, but its datasets, algorithm, and statistical testing differ from the present implementation [1].

6.1 Interpretation in the Context of Prior Work

The current results should be positioned as a controlled project experiment rather than a leaderboard comparison. The reviews by Ngo *et al.*, Amato *et al.*, and Xavier *et al.* cover different study periods and inclusion strategies, but converge on a diverse methodological landscape [2–4]. The 2025 review by Farajollahi *et al.* also identifies gaps in reporting and validation that affect translation from research datasets to clinical settings [5]. This makes participant-disjoint testing and explicit disclosure of the test cohort important, but a single held-out split is not a substitute for independent external validation.

The present study's 65-person cohort matches the participant counts described for IPVS by Hossain *et al.* [6]. That correspondence supports the dataset description, but it does not mean the experiments are directly comparable. Differences in feature definitions, data partitions, model selection, and scoring units can materially change reported performance. The supplied ISNDAM paper uses a different algorithmic approach and reports results on its own experimental setup [1]. Accordingly, this paper compares only the two models implemented in the project notebook and avoids claiming improvement over published methods.

7 Limitations and Future Work

The experiment uses one dataset and one saved participant-level train/test split, with only 17 held-out participants. The healthy-control class combines young and elderly cohorts; age and cohort composition may therefore influence the acoustic patterns. Recording devices, environments, tasks, language, and speaking styles may also affect the signal. The CNN uses only the first five seconds of each recording, potentially omitting useful information later in the utterance.

The notebook does not report external-dataset validation, prospective clinical evaluation, probability calibration, or repeated independent test splits. Future work should evaluate additional participants and independent datasets, use repeated grouped or nested grouped validation, test different segment-selection strategies, assess recording-condition robustness, and examine calibration and uncertainty. Clinical application would require prospective validation and expert oversight.

8 Conclusion

This project implemented two participant-aware voice-classification pipelines for the IPVS collection: a classical pipeline using 115 handcrafted acoustic feature columns and a CNN using five-second Log-Mel spectrograms. The RBF-SVM achieved 0.9558 recording-level accuracy and 0.9412 participant-level accuracy. The CNN achieved 0.9690 recording-level accuracy and the same participant-level accuracy. The models therefore show promising discrimination within the saved experiment, but the small held-out participant cohort, single dataset, and single split

prevent broad generalisation claims. The work is an experimental research prototype and not a clinical diagnostic system.

Author Contributions

Suvinay Foujdar: project coordination, integration of the experimental pipeline, experimental design, and manuscript preparation. Akshat Goyal: dataset organisation, audio preprocessing, handcrafted-feature extraction, and classical machine-learning experiments. Navya Loshali: literature review, Log-Mel CNN experiments, result visualisation, and manuscript review. The listed roles are provisional and should be confirmed by all authors before submission.

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References

[1] I. Luna-Ortiz, M. Aldape-Pérez, A. V. Uriarte-Arcia, A. Rodríguez-Molina, A. Alarcón-Paredes, and E. Ventura-Molina, "Parkinson's Disease Detection from Voice Recordings Using Associative Memories," *Healthcare*, vol. 11, no. 11, Art. no. 1601, 2023, doi: https://doi.org/10.3390/healthcare11111601.

[2] Q. C. Ngo, M. A. Motin, N. D. Pah, P. Drotár, P. Kempster, and D. Kumar, "Computerized analysis of speech and voice for Parkinson's disease: A systematic review," *Computer Methods and Programs in Biomedicine*, vol. 226, Art. no. 107133, 2022, doi: https://doi.org/10.1016/j.cmpb.2022.107133.

[3] F. Amato, G. Saggio, V. Cesarini, G. Olmo, and G. Costantini, "Machine learning- and statistical-based voice analysis of Parkinson's disease patients: A survey," *Expert Systems with Applications*, vol. 219, Art. no. 119651, 2023, doi: https://doi.org/10.1016/j.eswa.2023.119651.

[4] D. Xavier, V. Felizardo, B. Ferreira, H. Zacarias, M. Pourvahab, L. Souza-Pereira, and N. M. Garcia, "Voice analysis in Parkinson's disease—a systematic literature review," *Artificial Intelligence in Medicine*, vol. 163, Art. no. 103109, 2025, doi: https://doi.org/10.1016/j.artmed.2025.103109.

[5] B. Farajollahi, A. Saffarian, S. A. H. Habibi, and A. Sheikhtaheri, "Voice-Based Detection of Parkinson's Disease Using Machine and Deep Learning Approaches: A Systematic Review," *Bioengineering*, vol. 12, no. 11, Art. no. 1279, 2025, doi: https://doi.org/10.3390/bioengineering12111279.

[6] M. A. Hossain, E. Traini, and F. Amenta, "Identification of Parkinson's Disease from Native Italian People: Machine Learning Voice Analysis," *BioMed*, vol. 6, no. 3, Art. no. 15, 2026, doi: https://doi.org/10.3390/biomed6030015.